

ESBJERG SAEDENSTRAND, Denmark

WMO: 060890

Lat: 55.500N

Lon: 8.400E

Elev: 36

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 82-92

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.2	18.9	9.9	9.2	14.1	15.2	11.9	20.1	50.7	42.3	47.5	43.2	9.7	20	0.838

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	77.6	67.2	73.3	64.8	69.7	63.0	68.4	75.9	66.1	71.2	64.0	67.9	12.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	94.5	71.1	63.9	88.9	69.2	62.1	83.6	66.9	32.6	75.9	30.8	71.0	29.3	68.2	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 42.7	34.3	30.9	13.7	83.2	8.5	4.3	7.6	86.4	2.6	88.9	-2.1	91.4	-8.3	94.5		
(5)			13.2	71.4	8.4	2.2	7.2	72.9	2.3	74.2	-2.4	75.4	-8.5	77.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	34.6	33.8	37.9	43.5	52.2	57.0	61.7	61.4	56.3	50.8	42.2	38.1		
	DBStd	11.23	8.48	6.75	5.22	4.76	5.47	5.01	4.45	3.94	3.46	4.30	5.58	4.81		
	HDD50	2194	478	454	377	204	34	1	0	0	1	42	235	368		
	HDD65	6417	943	874	841	644	397	247	126	126	262	441	683	833		
	CDD50	1298	0	0	0	10	103	211	362	354	190	66	2	0		
	CDD65	48	0	0	0	0	2	7	23	15	1	0	0	0		
	CDH74	316	0	0	0	0	13	52	157	85	9	0	0	0		
	CDH80	58	0	0	0	0	0	10	28	19	1	0	0	0		
	WSAvg	16.0	17.9	16.1	16.3	14.3	14.4	14.8	16.3	16.1	16.7	16.8	15.3	16.3		
	PrecAvg	30.8	2.4	1.6	1.9	1.5	1.7	2.1	2.5	2.8	3.4	3.8	3.9	3.1		
	PrecMax	43.6	5.7	4.9	4.6	3.1	7.6	5.3	6.9	6.7	8.0	9.4	6.9	5.9		
	PrecMin	20.5	0.2	0.0	0.2	0.0	0.4	0.2	0.5	0.1	0.5	0.9	1.7	0.4		
	PrecStd	5.4	1.1	1.2	1.1	0.8	1.3	1.2	1.4	1.5	1.9	2.0	1.3	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	46.6	53.0	64.0	76.0	80.9	84.2	82.8	72.1	62.7	54.1	49.3		
		MCWB	45.0	45.6	48.3	53.3	63.9	66.2	70.5	69.3	65.5	59.6	52.3	48.2		
	2%	DB	44.9	45.1	48.4	59.3	70.8	73.6	78.3	75.3	65.4	59.6	51.2	46.3		
		MCWB	43.9	43.9	46.1	51.6	59.3	63.7	67.8	67.4	60.1	57.4	50.0	45.0		
	5%	DB	44.0	43.8	45.9	54.0	66.3	69.6	74.1	70.2	63.0	57.5	50.0	44.9		
		MCWB	43.1	42.6	44.4	49.2	57.3	62.0	65.6	64.9	59.3	55.4	48.9	43.9		
	10%	DB	43.3	42.5	44.3	50.6	61.8	65.4	69.4	67.2	61.4	55.9	48.9	44.1		
		MCWB	42.4	41.4	43.1	46.9	55.2	59.8	63.7	63.2	58.3	54.3	47.8	43.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	45.9	49.6	55.0	65.1	68.7	72.0	70.7	65.4	60.5	52.9	48.4		
		MCDB	45.7	46.2	52.7	61.5	74.1	78.0	80.6	78.8	70.2	62.0	53.5	49.3		
	2%	WB	44.3	44.2	46.4	52.4	61.2	65.1	68.9	68.0	61.9	58.0	50.3	45.6		
		MCDB	44.6	44.8	47.8	57.9	67.8	70.8	77.0	74.8	63.7	59.0	50.9	46.0		
	5%	WB	43.5	43.0	44.7	49.8	58.4	62.6	66.3	65.8	60.3	56.1	49.2	44.2		
		MCDB	43.9	43.7	45.6	53.8	64.2	68.2	72.0	69.6	61.9	56.9	49.8	44.7		
	10%	WB	42.6	41.4	43.3	47.5	55.7	60.4	64.1	63.8	58.9	54.8	48.0	43.4		
		MCDB	43.1	42.1	44.2	50.2	60.8	64.8	68.3	66.6	60.7	55.5	48.6	43.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.4	5.8	6.2	9.4	10.9	9.8	9.6	8.7	7.7	6.9	6.3	5.3		
		MCDBR	4.2	4.6	9.1	16.2	20.2	18.5	20.2	15.9	10.7	8.9	6.1	5.0		
		MCWBR	4.3	4.5	7.2	10.2	11.4	10.6	10.8	9.7	7.0	6.5	5.5	5.0		
	5% WB	MCDBR	4.2	4.0	7.7	14.9	18.1	16.2	17.6	14.4	8.8	7.5	5.7	4.7		
		MCWBR	4.5	4.3	6.5	9.8	11.2	10.4	10.4	9.5	7.2	6.4	5.5	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.342	0.364	0.397	0.388	0.385	0.398	0.400	0.380	0.359	0.336	0.312		
	taud		2.248	2.300	2.301	2.259	2.350	2.380	2.359	2.361	2.399	2.380	2.301	2.246		
	Ebn at Noon		182	228	253	260	271	273	266	258	246	220	178	158		
	Edh at Noon		18	25	31	37	36	36	36	34	28	23	17	15		
(44) All-Sky Solar Radiation	RadAvg		155	351	771	1290	1687	1779	1664	1334	918	474	193	114		
	RadStd		26	49	71	148	164	138	173	95	94	57	29	12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-211	-290	N/A

Nomenclature: See separate page